PARTNERSHIP MEMO

Korea–Canada Strategic Energy & Industrial Partnership

Where deeper industrial integration creates strategic value, through which instruments, and where to stay out

Generated from Allied Industrial Base Explorer data and a curated external evidence layer (accessed 2026-11-06); evidence, sources and caveats in the appendix. Data as of 30 October 2026.

Executive Decision Summary

BUILD TOGETHER

Permanent Magnets & Drive Systems: Electric Vehicles refining (joint). Nuclear Fuel & Reactor Systems: Uranium conversion & enrichment (Canada->Korea). Battery Materials & Cells: Accumulator parts & separators (Korea->Canada). Grid Equipment & Conductors: Solar refining (joint).

TRADE AND STANDARDS

Recognise trusted partner supply of batteries extraction (cobalt ores, lithium minerals, natural graphite); accumulator parts & separators; oil extraction in procurement and content rules. Common anti-circumvention rules for electric vehicles refining (rare-earth metals, rare-earth oxides & compounds). Keep nuclear special materials; battery recycling feed; semiconductor manufacturing equipment open to global sourcing.

INSTITUTIONS

Upgrade Critical-minerals and energy-security MOU (2023) and 2026 stockpiling track: sector strategy, reciprocal commitments, a KPI dashboard and an annual review. No new MOU.

STAY OUT

Do not co-subsidise new Batteries assembly (li-ion cells & packs, other accumulators) capacity. Do not create a new bilateral energy MOU or a free-standing MEA institution. Do not put new public money behind Cathode materials & precursors investment that is already committed (evidence KRCA-17.

Why this partnership matters.

- Of 142 technology x stage cells and electro-industrial nodes assessed for the pair, 22 show an active complementarity (a supplier meeting the other's need, a shared chokepoint, a capacity-constrained trusted supplier or a chain specialisation), 1 direct competition and 94 no strategic case. The relationship is not one of general similarity: most of the electro-industrial base needs no bilateral intervention.[1]
- AIB strategic roles: Korea Specialized Technology & Manufacturing Anchor (primary), Scale Manufacturing & Integration Platform (primary), Coordination / Bridge Hub (secondary), Capital & Investment Anchor (secondary); Canada Resource & Processing Anchor (primary), Capital & Investment Anchor (secondary), Demand & Market Anchor (secondary).[1]

The central bilateral bargain.

- *Inference.* Canada supplies uranium conversion & enrichment, nickel intermediates & chemicals, aluminium wire rod where Korea is short; Korea supplies accumulator parts & separators, cathode materials & precursors, non-oriented electrical steel (ngo), power cables where Canada is short; they share no node-level chokepoint where neither supplies.

- *Inference.* From the AIB roles, Korea brings specialised technology and manufacturing, scale manufacturing, coordination across allied chains; Canada brings resources and processing, capital, demand and offtake. Public finance runs through KEXIM, K-SURE, KOMIR, KDB on the Korea side and ISED, EDC, CIB, CGF, CMIF, FinDev on the Canada side. The trade is market access and demand certainty for trusted capacity.

The 4 highest-priority joint bets (tier 1 by the rules in Section 3, one per compact; ranked by urgency then importance):

PRIORITY	SECTOR	NODES IN THE BET (DIRECTION)	STRATEGIC PROBLEM	KOREA CONTRIBUTION	CANADA CONTRIBUTION	MAIN INSTRUMENT (LEAD NODE)	TIMING
1	Permanent Magnets & Drive Systems	Electric Vehicles refining (joint)	geopolitical chokepoint	Electric Vehicles refining: its demand for pooled offtake, and finance for non-Chinese supply; Wind refining: its demand for pooled offtake, and finance for non-Chinese supply	Electric Vehicles refining: its demand for pooled offtake, and finance for non-Chinese supply; Wind refining: its demand for pooled offtake, and finance for non-Chinese supply	anchor offtake	12-24 months
2	Nuclear Fuel & Reactor Systems	Uranium conversion & enrichment (Canada->Korea)	strategic manufacturing gap	Uranium conversion & enrichment: demand and long-term offtake; Nuclear refining: demand and long-term offtake	Uranium conversion & enrichment: trusted capacity (16% of world supply); Nuclear refining: trusted capacity (top-quarter position)	strategic imports	12-24 months
3	Battery Materials & Cells	Accumulator parts & separators (Korea->Canada)	trusted partner capacity constraint	Accumulator parts & separators: trusted capacity (13% of world supply); Batteries extraction: demand and long-term offtake	Accumulator parts & separators: demand and long-term offtake; Batteries extraction: trusted capacity (top-quarter position)	anchor offtake	12-24 months
4	Grid Equipment & Conductors	Solar refining (joint)	geopolitical chokepoint	Solar refining: its demand for pooled offtake, and finance for non-Chinese supply	Solar refining: its demand for pooled offtake, and finance for non-Chinese supply	anchor offtake	12-24 months

Sources: 1.

- *Inference.* Outside the electro-industrial compacts, the rules also find fuel-trade lanes (oil extraction; coal components; coal assembly); they are handled in the trade agenda, not as compacts.

Do not prioritise (Section 10):

- *Recommendation.* Do not co-subsidise new Batteries assembly (li-ion cells & packs, other accumulators) capacity: both are strong suppliers (Korea 5%, Canada 1%). More trusted capacity is needed, so agree who specialises in what rather than racing.
- *Recommendation.* Do not create a new bilateral energy MOU or a free-standing MEA institution: KRCA-M2 already exists (a standing framework with no pipeline or finance yet); dock the compacts in it.
- *Recommendation.* Do not put new public money behind Cathode materials & precursors investment that is already committed (evidence KRCA-17; KRCA-18); support is additional only beyond it.

Existing mechanisms and the MEA question (Section 8):

- *Inference.* MEA assessment: **upgrade an existing framework (KRCA-M2)**. Gaps: no cross-sector KPI dashboard or review cycle; KRCA-M2 has no published project pipeline.

- *Recommendation.* Do not sign a new energy MOU. Give KRCA-M2 a sector strategy, reciprocal commitments and a KPI dashboard covering the recommended compacts, reviewed every 12 months.

1. Current State of the Relationship

This section sets out what already exists before anything new is proposed. Facts are from the curated evidence layer (accessed 2026-11-06); items marked partial or unverified rest on secondary sources and should be confirmed before citation.

Alliance and plurilateral architecture

- Canada's memberships in OSI's economic-security agreements table: G7, MSP, NATO, U.S. FTA.[1]
- Economic-security 2+2 (Joly, Champagne, Park Jin, Ahn Dukgeun) launched Sep 2022 and first met in Seoul on 16 May 2023; priorities were critical minerals, clean energy, supply chains, AI and economic coercion.[2]
- Second 2+2 (first was Nov 2024) agreed to pursue an update of the Comprehensive Strategic Partnership Action Plan, start Defence Cooperation Agreement steps and coordinate on economic security; the statement does not mention critical minerals or energy; next meeting in Korea in 2028.[3]
- On 2 Jun 2026 Canada and Korea agreed to develop a joint critical-minerals stockpiling plan by end-2026, discussed strategic-resource investments, and Korea agreed to facilitate CKFTA preferential tariffs for Canadian crude; no volumes or amounts given.[4]
- Lee and Carney agreed on 16 Jun 2026 to strengthen cooperation in defence, security, energy and critical minerals; an informal meeting followed at the NATO summit; no deliverables or amounts reported.[5]

Trade architecture and frictions

- In the OSI energy basket, Korea imported US$3.7bn from Canada and Canada imported US$3.4bn from Korea (bilateral flows 2024, summed over the technology x stage cells).[1]
- Signed in Ottawa 22 Sep 2014 and in force 1 Jan 2015; Canada's first FTA in Asia-Pacific, covering goods, services, investment and procurement; nearly 90% of Canadian exports to Korea duty-free at entry into force.[6]
- From 26 Dec 2025 Canada cut FTA-partner steel quotas to 75% of 2024 volumes with a 50% over-quota surtax; Korea protested that this breaches the CKFTA and sought an exemption, none found.[7]

Industrial and energy agreements

- The Comprehensive Strategic Partnership was launched in 2022 and is the base for the Jan 2026 industrial MOU; leaders agreed in Sep 2022 to develop a critical-minerals and energy MOU; day of launch inferred from the Sep 2022 summit.[8]
- Non-binding MOU (2023, not 2022) covering critical-mineral value chains, clean fuels and CCS, and clean-energy, ZEV and semiconductor manufacturing, with an action plan and investment-screening cooperation; it lets Korea count Canada toward the US EV-credit minerals test.[9]
- Carney and Lee welcomed the partnership at the Gyeongju APEC summit on 30 Oct 2025, Canada's first in the Indo-Pacific, and finalised a classified-information agreement that supports defence procurement; the readout does not mention energy, critical minerals or LNG.[10]
- Joly and Korea's Trade, Industry and Resources Minister Kim Jung-kwan signed in Ottawa; sets up an industrial cooperation committee and a forum on a Korean auto footprint, battery production, materials processing and critical-minerals recycling; no amounts or companies named.[11]
- The Geological Survey of Canada and Korea's KIGAM signed an implementation agreement on naturally occurring hydrogen research and development, announced with the 2 Jun 2026 energy and critical-minerals meeting.[12]

Investment vehicle, project pipeline and financing

Corporate relationships in the recommended compacts

- Samsung SDI bought 15.6m Canada Nickel shares for US$18.5m with an option on 10% of the Crawford nickel-cobalt sulphide project carrying offtake rights; date not shown in the source.[13]
- In 2022 LGES signed non-binding MOUs for 7,000 t of cobalt sulphate over three years from 2023 and 55,000 t of lithium hydroxide over five years from 2025; signing date inferred from the Sep 2022 Toronto visit.[14]
- C$1.2bn, 45,000 t/yr plant (originally 1H 2026) was halted twice in 2024 and resumed in Sep 2024; one undated Korean report says Ford is leaving; USD figure is the 2023 announcement value.[15]
- JV owned 85% by POSCO Future M and 15% by GM; phase 1 (20,000 t/yr high-nickel cathode, all for Ultium Cells) was scheduled for late Oct 2026 after earlier delays; phase 2 timing unconfirmed; no source found confirming start-up has occurred.[16]
- LGES agreed in early Feb 2026 to take full ownership of NextStar by buying Stellantis's 49%, with Stellantis staying a customer; publication day approximate.[17]
- Series cell production began Nov 2025 and the plant was officially opened in Mar 2026, initially focused on stationary storage; employment 1,300+ against a 2,500 target.[18]
- Binding agreement for 100% of Phase 1 (about 150 t/yr LCE) for ten years, renewable to 30; crystallite reported to be refined in Korea, first production targeted Q4 2026; sources differ on whether Hydro Lithium is Korean-based.[19]
- Unverified items are left out of this section; they stay in the evidence register (evidence_register.csv) and the evidence layer.[1]
- *Inference.* Working channels today: Industrial cooperation MOU and Canada-Korea Industrial Cooperation Committee (Jan 2026) (a project pipeline). What the relationship lacks: no cross-sector KPI dashboard or review cycle; KRCA-M2 has no published project pipeline.

2. What Each Country Is Good At — and What It Needs

Cell codes: position / need for **Korea · Canada**. Position S strong (top quarter of contenders and above the capability floor), M contender, W not a contender; need H high (supply-chain need >= 0.67, or deployment need >= 0.60 at Deployment), L low; - not measured. Source: strategic_position_tbl (engine choices are in country_profiles.csv).

TECHNOLOGY	EXTRACTION	REFINING	COMPONENTS	ASSEMBLY	DEPLOYMENT	RECOVERY
Batteries	W/H · M/H	W/H · W/L	M/H · M/H	S/H · S/H	S/L · W/L	-/H · -/H
Coal	W/H · M/L	W/H · W/H	S/L · M/H	S/L · M/H	S/L · S/L	-/- · -/-
Electric Grid	W/H · W/H	W/H · W/L	M/H · M/L	M/L · M/L	-/L · -/L	-/- · -/-
Electric Vehicles	W/H · M/H	W/H · M/H	S/H · M/L	S/L · M/L	S/H · S/L	-/- · -/-
Gas	W/H · M/H	W/H · M/H	M/L · M/L	S/L · M/L	S/L · S/H	-/- · -/-
Geothermal	-/- · -/-	-/- · -/-	-/- · -/-	-/- · -/-	M/H · S/L	-/- · -/-
Green Hydrogen	W/- · M/-	W/- · S/-	-/- · -/-	-/- · -/-	-/H · -/L	-/- · -/-
Nuclear	W/H · S/L	W/H · M/L	S/L · M/H	S/L · M/H	S/H · S/L	-/- · -/-
Oil	W/H · S/H	M/L · M/L	M/H · M/L	S/L · M/L	S/L · S/L	-/L · -/H
Solar	W/H · W/H	W/H · W/H	M/L · M/L	M/L · M/L	M/H · S/L	-/- · -/-
Wind	W/- · W/-	W/H · W/H	M/H · M/L	M/L · M/L	S/H · S/L	-/- · -/-

Sources: 1.

Electro-industrial nodes where either country has a position or a need (AIB division of labour, 2024 trade; share = world export or production share; need is the node's technology x stage need, inherited, so a proxy; demand = share of allied demand). 62 of 76 nodes:

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

Table A1 in the appendix (62 rows) carries this table in full; the CSV outputs carry every column.

- Revealed policy priority (NIPO / IEA PAMS within-country percentile; policy never enters the classification) is highest for Korea in: Batteries assembly (0.96); Green Hydrogen assembly (0.95); Nuclear assembly (0.92); Nuclear components (0.90); Electric Vehicles recovery (0.87); for Canada in: Gas extraction (0.98); Electric Vehicles extraction (0.90); Gas components (0.87); Green Hydrogen assembly (0.86); Oil extraction (0.84).[1]
- Evidence quality: 39 of 66 cells have high stage-specific evidence for both countries; 19 units are flagged low-evidence.[1]

3. Bilateral Complementarity Map

State counts over all units: a_supplies_b_needs 4; b_supplies_a_needs 4; complementary_specialization 5; shared_vulnerability 4; capacity_constrained_trusted_supply 5; direct_competition 1; low_strategic_case 94; deployment_demand 7; not_assessed 18.[1]

Inference. Rules (Section 3 of docs/partnership_memos.md): two strong positions are **direct competition**, never complementarity; a supplier must be strong (or a contender the engine names as the other's trusted partner) and the buyer must have high need, or material demand where trusted capacity is short or a chokepoint exists; **capacity-constrained** where the AIB finds allied capacity short of demand; **shared vulnerability** where both need the node, neither supplies it and a coercive supplier holds at least half of world supply.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

Table A2 in the appendix (23 rows) carries this table in full; the CSV outputs carry every column.

- Low-priority / open-market nodes (69): wafers, polysilicon, synthetic graphite (anode-grade), solar cells, pv modules & generators, li-ion cells & packs, rare-earth oxides & compounds, rare-earth metals, natural graphite, nuclear reactors & enrichment machinery, fuel assemblies & zirconium, reactor parts, uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores), electric motors, rare-earth permanent magnets, electric buses, trucks & tractors, electric & plug-in cars, electric two-wheelers & works trucks, ferrite & other magnets, motor & generator parts, rare-earth ores (monazite), cobalt intermediates & chemicals, wind turbine generating sets, lithium chemicals, pv cover glass, wind blades & hubs, wind towers, piezo-electric quartz, copper smelting & refining, refrigerants, heat pumps & ac units, manganese dioxide (emd), static converters & inverters, power electronics components, solar thermal water heaters, gallium, germanium & indium, grain-oriented electrical steel (goes), electrolysers, refrigeration & heat-pump compressors, power semiconductors, sensors & controller parts, insulators & bushings, other accumulators, electricity meters, battery materials (foil, electrolyte), electric furnaces & heating, furnace parts, lithium minerals, battery recycling feed, automatic controllers, integrated circuits, transformers, switchgear & protection, hydro turbine parts, graphite electrodes, blade core wood (balsa), ac generators, copper ores & concentrates, semiconductor manufacturing equipment, hydrogen, ammonia & methanol, nuclear special materials, silver powder, direct-reduced iron (dri/hbi), cobalt ores, electric rail vehicles, overhead conductors, copper wire & rod, hydro turbines, pgms (iridium, ruthenium).[1]
- Deployment cells are demand context, not supply relations: high deployment need in Green Hydrogen (KOR); Nuclear (KOR); Electric Vehicles (KOR); Solar (KOR); Wind (KOR); Geothermal (KOR); Gas (CAN).[1]

4. Trade Agenda

- *Inference.* **What Korea should export to Canada:** Nuclear components (fuel assemblies & zirconium, reactor parts); Nuclear assembly (nuclear reactors & enrichment machinery); Accumulator parts & separators; Coal components; Coal assembly; Cathode materials & precursors; Non-oriented electrical steel (NGO); Power cables; Oil assembly (ac generators).
- *Inference.* **What Canada should export to Korea:** Batteries extraction (cobalt ores, lithium minerals, natural graphite); Nuclear refining (uranium conversion & enrichment); Uranium conversion & enrichment; Nuclear extraction (uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores)); Oil extraction; Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite); Nickel intermediates & chemicals; Aluminium wire rod; Coal extraction.
- *Inference.* **Where tariffs would undermine allied specialisation:** no node where the buyer's protection priority is in its top third and the partner is the trusted supplier.
- *Inference.* **Procurement, standards and content rules to recognise mutually:** Batteries|Extraction, accumulator-parts-and-separators, Oil|Extraction.
- *Inference.* **Where third-country circumvention safeguards are genuinely required:** Electric Vehicles|Refining; elsewhere they would add cost without security.
- *Inference.* **Where unrestricted global sourcing is acceptable:** nuclear-special-materials, battery-recycling-feed, semiconductor-manufacturing-equipment, copper-ores-and-concentrates, lithium-minerals, silver-powder.

Table A3 in the appendix (19 rows) carries this table in full; the CSV outputs carry every column.

- The tariff measures in force or in dispute between the two, the baseline any binding of the nodes above works from:
- From 26 Dec 2025 Canada cut FTA-partner steel quotas to 75% of 2024 volumes with a 50% over-quota surtax; Korea protested that this breaches the CKFTA and sought an exemption, none found.[7]

5. Investment Agenda

Korea → Canada investment (Korea capital and firms expanding in Canada)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I03	Nuclear extraction (uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores))	KOR	CAN	supply-chain need 0.96 (KOR) / 0.61 (CAN)		strategic imports	medium	2-3 years
I04	Nuclear components (fuel assemblies & zirconium, reactor parts)	KOR	CAN	supply-chain need 0.54 (KOR) / 0.74 (CAN)		export credit	medium	2-3 years
I05	Nuclear assembly (nuclear reactors & enrichment machinery)	KOR	CAN	supply-chain need 0.62 (KOR) / 0.67 (CAN)		export credit	medium	2-3 years
I06	Accumulator parts & separators	KOR	CAN	CAN takes 5% of allied demand (5% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I07	Oil extraction	KOR	CAN	supply-chain need 0.99 (KOR) / 0.84 (CAN)		anchor offtake	high	2-3 years
I08	Coal components	KOR	CAN	supply-chain need 0.58 (KOR) / 0.81 (CAN)		export credit	medium	2-3 years
I09	Coal assembly	KOR	CAN	supply-chain need 0.60 (KOR) / 0.68 (CAN)		export credit	medium	2-3 years
I10	Cathode materials & precursors	KOR	CAN	CAN takes 1% of allied demand (0% of allied net imports); AIB class BUILD	POSCO Future M; General Motors; EcoPro BM	export credit	low	2-3 years
I11	Non-oriented electrical steel (NGO)	KOR	CAN	CAN takes 4% of allied demand (4% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years

Sources: 1.

Canada → Korea investment (Canada capital and firms expanding in Korea)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I01	Nuclear refining (uranium conversion & enrichment)	CAN	KOR	supply-chain need 0.98 (KOR) / 0.61 (CAN)		export credit	medium	2-3 years
I02	Uranium conversion & enrichment	CAN	KOR	KOR takes 10% of allied demand (10% of allied net imports); AIB class LOW INTERVENTION		export credit	medium	2-3 years

Sources: 1.

Joint investment (at home or in third countries)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I12	Batteries extraction (cobalt ores, lithium minerals, natural graphite)	KOR+CAN	third countries (CHL; ARG; BRA) and either partner	supply-chain need 1.00 (KOR) / 0.96 (CAN)		anchor offtake	high	2-3 years
I13	Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite)	KOR+CAN	third countries (CHL; ARG; BRA) and either partner	supply-chain need 1.00 (KOR) / 0.96 (CAN)	Arafura Rare Earths	anchor offtake	low	2-3 years

Sources: 1.

- *Inference.* Additionality = mean of three named components: the market failure (chokepoint 1, capacity constraint 0.75, gap or emerging bottleneck 0.5, none 0), one minus an existing commitment in the evidence

layer (committed 1, announced 0.5) and one minus a working market (the buyer already takes 30% or more of its imports from the supplier). It is **low** whenever there is no failure or the investment is already committed: public support would then not be additional.

6. Third-Country and Development Cooperation

- From the development-partnership pairs (scripts/93: a donor's need meets a developing economy's credible Catch Up pathway in the same step), Korea and Canada are both donors at 10 recipient steps. Ranked: both donors' own need first, then steps inside a recommended compact.[1]
- *Inference.* Roles: Korea brings specialised technology and components and equipment and manufacturing at scale; Canada brings resources and processing and capital and export finance (AIB strategic roles).

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	RECIPIENT	TECHNOLOGY	STAGE	COMPACT	RECIPIENT CHOICE	RECIPIENT ANCHOR	KOREA INTEREST	CANADA INTEREST
X01	Viet Nam	Batteries	Components	SC-BATTERIES	catch up	direct supply share; direct export share	own catch up need (KOR)	own catch up need (CAN)
X02	Viet Nam	Batteries	Assembly	SC-BATTERIES	catch up	direct supply share; direct export share	own preserve need (KOR)	own preserve need (CAN)
X03	Indonesia	Batteries	Components	SC-BATTERIES	catch up	direct export share	own catch up need (KOR)	own catch up need (CAN)
X04	Malaysia	Batteries	Assembly	SC-BATTERIES	catch up	direct supply share; direct export share	own preserve need (KOR)	own preserve need (CAN)
X05	Indonesia	Batteries	Assembly	SC-BATTERIES	catch up	direct supply share	own preserve need (KOR)	own preserve need (CAN)
X06	Mexico	Batteries	Assembly	SC-BATTERIES	catch up	direct supply share; direct export share	own preserve need (KOR)	own preserve need (CAN)
X07	Mexico	Batteries	Components	SC-BATTERIES	catch up	direct supply share; direct export share	own catch up need (KOR)	own catch up need (CAN)
X08	Turkiye	Batteries	Components	SC-BATTERIES	catch up	direct supply share; direct export share	own catch up need (KOR)	own catch up need (CAN)
X09	India	Batteries	Assembly	SC-BATTERIES	catch up	committed pipeline	own preserve need (KOR)	own preserve need (CAN)
X10	India	Batteries	Components	SC-BATTERIES	catch up	committed pipeline	own catch up need (KOR)	own catch up need (CAN)

Sources: 1.

- Final investment decision in May 2026 after an Australian offtake commitment under the Strategic Reserve; construction began 8 Oct 2026; credit agencies from the US, Canada, Korea and Germany support it; the article's A$1.2bn figure is ambiguous.[20]

7. Recommended Sectoral Compacts

Inference. 1 compacts pass (rule: at least one joint bet, at least two active nodes and both countries holding an AIB compact role; or, labelled **evidence-led**, an existing bilateral pipeline or at least 2 bilateral project items where OSI measures a need but cannot see the partner's capability). The hypothesis test for every AIB compact:

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

RANK	COMPACT	VERDICT	JOINT BETS	COORDINATE	KOREA ROLE	CANADA ROLE	BILATERAL MECHANISMS	RECOMMENDED
1	Battery Materials & Cells	bilateral priority	1	2	Specialized technology supplier	Processing anchor	KRCA-M2; KRCA-M3	TRUE
2	Nuclear Fuel & Reactor Systems	secondary	1	0	Scale manufacturer	Processing anchor		FALSE
3	Permanent Magnets & Drive Systems	secondary	0	1	Specialized technology supplier			FALSE
4	Grid Equipment & Conductors	secondary	0	2	Scale manufacturer	Specialized technology supplier		FALSE
5	Industrial Electrification & Hydrogen	not a bilateral priority	0	0	Scale manufacturer			FALSE
6	Electric Vehicles	not a bilateral priority	0	0	Scale manufacturer	Scale manufacturer	KRCA-M3	FALSE
7	Heat Pumps & Cooling	not a bilateral priority	0	0	Scale manufacturer			FALSE
8	Semiconductors & Power Electronics	not a bilateral priority	0	0	Specialized technology supplier			FALSE
9	Solar Manufacturing	not a bilateral priority	0	0	Specialized technology supplier			FALSE
10	Wind & Hydropower Equipment	not a bilateral priority	0	0	Specialized technology supplier	Diversification partner		FALSE

Sources: 1.

1. Battery Materials & Cells

Strategic objective

- AIB mission: Build missing capacity in Li-ion cells & packs and lithium chemicals and manganese dioxide (EMD) (refined chemicals); targeted: preserve and second-source specialized production in lithium chemicals (refined chemicals)..[1]
- *Recommendation.* Change the outcome at accumulator parts & separators, cathode materials & precursors, nickel intermediates & chemicals: trusted partner capacity constraint; geopolitical chokepoint; strategic manufacturing gap.

Why this pair?

- *Inference.* Accumulator parts & separators: Capacity-constrained trusted supply (KOR is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Cathode materials & precursors: Complementary specialisation (KOR is strong here and the other country is strong at another step of the same chain). Nickel intermediates & chemicals: Complementary specialisation (CAN is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: Korea Specialized technology supplier (lead node cathode-materials-and-precursors); Canada Processing anchor (lead node nickel-intermediates-and-chemicals).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	KOREA POSITION	CANADA POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	natural-graphite	AIB node	moderate	moderate	low strategic case	LOW INTERVENTION	
Refining	cathode-materials-and-precursors	AIB node	strong	moderate	complementary specialization	COORDINATE	standards alignment
Components	accumulator-parts-and-separators	AIB node	strong	weak	capacity constrained trusted supply	BUILD	anchor offtake
Assembly	li-ion-cells-and-packs	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Deployment	Batteries/Deployment	OSI cell (no AIB node at this stage)	strong	weak	low strategic case	LOW INTERVENTION	
Recovery	battery-recycling-feed	AIB node	weak	moderate	low strategic case	LOW INTERVENTION	

Sources: 1.

Reciprocal bargain

- Korea provides: supply of accumulator parts & separators, cathode materials & precursors; long-term demand and offtake for nickel intermediates & chemicals; procurement recognition of the partner as trusted supply; finance for the compact's nodes (KEXIM, K-SURE, KOMIR, KDB).
- Canada provides: capacity and technology in nickel intermediates & chemicals; long-term demand for accumulator parts & separators, cathode materials & precursors; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (ISED, EDC, CIB, CGF, CMIF, FinDev).
- Jointly: common specifications and qualification.
- Third-country partners needed: JPN.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): DEU at synthetic graphite (anode-grade); POL at li-ion cells & packs; MDG at natural graphite; FIN at cobalt intermediates & chemicals; CHL at lithium chemicals; JPN at manganese dioxide (emd); VNM at other accumulators; TWN at battery materials (foil, electrolyte). The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* anchor_offtake for accumulator parts & separators; standards_alignment for cathode materials & precursors; standards_alignment for nickel intermediates & chemicals

Priority transactions

- *Recommendation.* Long-term offtake by Korea buyers for Canada batteries extraction (cobalt ores, lithium minerals, natural graphite), sized to underwrite new capacity (candidate hosts: CHL, ARG, BRA) (D01).

Success by 2030

- Accumulator parts & separators: allied residual gap under the China shock closed (now 37% of allied demand); Korea's share of Canada's imports up from 1%.
- Cathode materials & precursors: coercive share of supply to the two buyers below 50% (CHN now holds 92% of world supply); Korea's share of Canada's imports up from 0%.
- Nickel intermediates & chemicals: allied residual gap under the China shock closed (now 1% of allied demand); Canada's share of Korea's imports up from 2%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- Samsung SDI bought 15.6m Canada Nickel shares for US$18.5m with an option on 10% of the Crawford nickel-cobalt sulphide project carrying offtake rights; date not shown in the source.[13]
- In 2022 LGES signed non-binding MOUs for 7,000 t of cobalt sulphate over three years from 2023 and 55,000 t of lithium hydroxide over five years from 2025; signing date inferred from the Sep 2022 Toronto visit.[14]
- C$1.2bn, 45,000 t/yr plant (originally 1H 2026) was halted twice in 2024 and resumed in Sep 2024; one undated Korean report says Ford is leaving; USD figure is the 2023 announcement value.[15]
- JV owned 85% by POSCO Future M and 15% by GM; phase 1 (20,000 t/yr high-nickel cathode, all for Ultium Cells) was scheduled for late Oct 2026 after earlier delays; phase 2 timing unconfirmed; no source found confirming start-up has occurred.[16]
- Joly and Korea's Trade, Industry and Resources Minister Kim Jung-kwan signed in Ottawa; sets up an industrial cooperation committee and a forum on a Korean auto footprint, battery production, materials processing and critical-minerals recycling; no amounts or companies named.[11]
- LGES agreed in early Feb 2026 to take full ownership of NextStar by buying Stellantis's 49%, with Stellantis staying a customer; publication day approximate.[17]
- Series cell production began Nov 2025 and the plant was officially opened in Mar 2026, initially focused on stationary storage; employment 1,300+ against a 2,500 target.[18]
- Binding agreement for 100% of Phase 1 (about 150 t/yr LCE) for ten years, renewable to 30; crystallite reported to be refined in Korea, first production targeted Q4 2026; sources differ on whether Hydro Lithium is Korean-based.[19]

8. Master Energy Agreement Assessment

- *Inference.* Rule: **existing** if every recommended compact is covered by a mechanism with both a pipeline and finance and a cross-sector mechanism has KPIs and a review cycle; **upgrade** if at least 50% of compacts are covered by a mechanism with finance or a pipeline, or a cross-sector mechanism with a pipeline exists; **new MEA** otherwise.
- *Inference.* Result: **upgrade**; coverage 100% of recommended compacts; cross-sector mechanism with KPIs and review: no.

COMPACT	ANY MECHANISM	FINANCE OR PIPELINE	PIPELINE AND FINANCE
SC-BATTERIES	TRUE	TRUE	FALSE

Sources: 1.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	MECHANISM	STATUS	SCOPE	PIPELINE	FINANCE	KPIS	REVIEW	RECIPROCAL
KRCA-M1	Canada-Korea Free Trade Agreement (2015)	in force	cross sector	FALSE	FALSE	FALSE	partial	TRUE
KRCA-M2	Critical-minerals and energy-security MOU (2023) and 2026 stockpiling track	in force	SC-BATTERIES/cross sector	FALSE	FALSE	FALSE	partial	TRUE
KRCA-M3	Industrial cooperation MOU and Canada-Korea Industrial Cooperation Committee (Jan 2026)	signed	SC-BATTERIES/SC-EV	partial	FALSE	FALSE	TRUE	partial
KRCA-M4	Comprehensive Strategic Partnership, Security and Defence Partnership and 2+2 dialogues	in force	cross sector	FALSE	FALSE	FALSE	TRUE	TRUE

Sources: 1.

Recommended: upgrade Critical-minerals and energy-security MOU (2023) and 2026 stockpiling track into the execution framework for the compacts

- **Scope:** Battery Materials & Cells; no other sectors.
- **Reciprocal commitments:** Korea commits demand certainty (procurement recognition and anchor offtake) and market-access predictability for the nodes Canada supplies (nickel intermediates & chemicals); Canada commits the same for the nodes Korea supplies (accumulator parts & separators, cathode materials & precursors). Each commits public finance for the compact nodes, at home, in the partner or with named third-country hosts.
- **Compact portfolio:** Battery Materials & Cells (bilateral priority).
- **Financing lanes:** Korea: KEXIM, K-SURE, KOMIR, KDB; Canada: ISED, EDC, CIB, CGF, CMIF, FinDev; demand-side tools (offtake, stockpiles, procurement): MCEE, KOMIR, PPS, NSSC and NRCan, CNSC; third-country finance: KEXIM, KOMIR, KOICA and GAC, FinDev.
- **Governance:** one bilateral steering line inside Critical-minerals and energy-security MOU (2023) and 2026 stockpiling track, with the existing sector mechanisms running their workstreams (Industrial cooperation MOU and Canada-Korea Industrial Cooperation Committee (Jan 2026)). Projects are selected by a joint committee, not by one side.
- **Milestones:** the 12-month milestones of the deal pipeline (Section 9).
- **KPI dashboard:** per compact node, the coercive share of each country's supply, the bilateral import share, offtake signed (volume), capacity at FID and operating, and public money committed vs private.
- **Review cycle:** every 12 months, with the OSI tables re-run on new trade data.
- **Sunset / exit:** each compact carries the exit conditions in Section 7; the framework expires with the investment vehicle () unless renewed on evidence.

9. 12–36 Month Deal Pipeline

Inference. Ranked lexicographically on separate fields (importance, then additionality, urgency, tractability, execution speed, value-chain interdependency, evidence quality); urgency is not importance. At most two deals per compact.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

RANK	DEAL / ACTION	COMPACT	INSTRUMENT	RESPONSIBLE INSTITUTIONS	12M MILESTONE	IMPORTANCE	URGENCY	ADDITIONALITY
1	Long-term offtake by Korea buyers for Canada batteries extraction (cobalt ores, lithium minerals, natural graphite), sized to underwrite new capacity (candidate hosts: CHL, ARG, BRA)	SC-BATTERIES	anchor offtake	KOR: MCEE, PPS; CAN: NRCan	Signed long-term offtake from at least one large buyer	high	high	high
2	Long-term offtake by Korea and Canada buyers for non-coercive electric vehicles refining (rare-earth metals, rare-earth oxides & compounds), sized to underwrite new capacity (candidate hosts: THA, USA, MYS)	SC-MAGNETS	anchor offtake	KOR: MCEE, PPS; CAN: NRCan	Signed long-term offtake from at least one large buyer	high	high	high
3	Long-term offtake by Korea and Canada buyers for non-coercive solar refining (copper wire & rod, polysilicon), sized to underwrite new capacity (candidate hosts: MYS, JPN, DEU)	SC-GRID	anchor offtake	KOR: MCEE, PPS; CAN: NRCan	Signed long-term offtake from at least one large buyer	high	high	high
4	Long-term offtake by Korea buyers for Canada oil extraction, sized to underwrite new capacity		anchor offtake	KOR: MCEE, PPS; CAN: NRCan	Signed long-term offtake from at least one large buyer	high	low	high
5	Multi-year supply agreements for Korea nuclear components (fuel assemblies & zirconium, reactor parts) into Canada buyers	SC-NUCLEAR	strategic imports	KOR: MOTIR (MOTIE), K-SURE, KOTRA; CAN: GAC, EDC	Named import lanes and volumes agreed between buyers and the supplier's producers	high	high	medium
6	Multi-year supply agreements for Korea coal components into Canada buyers		strategic imports	KOR: MOTIR (MOTIE), K-SURE, KOTRA; CAN: GAC, EDC	Named import lanes and volumes agreed between buyers and the supplier's producers	high	low	medium

Sources: 1.

10. What Not to Do

- *Recommendation.* **N01** Do not co-subsidise new Batteries assembly (li-ion cells & packs, other accumulators) capacity: both are strong suppliers (Korea 5%, Canada 1%). More trusted capacity is needed, so agree who specialises in what rather than racing.
- *Recommendation.* **N02** Do not create a new bilateral energy MOU or a free-standing MEA institution: KRCA-M2 already exists (a standing framework with no pipeline or finance yet); dock the compacts in it.
- *Recommendation.* **N03** Do not put new public money behind Cathode materials & precursors investment that is already committed (evidence KRCA-17; KRCA-18); support is additional only beyond it.
- *Recommendation.* **N04** Do not put new public money behind Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite) investment that is already committed (evidence USAU-09); support is additional only beyond it.
- *Recommendation.* **N05** Do not put new public money behind Nickel intermediates & chemicals investment that is already committed (evidence KRCA-23); support is additional only beyond it.
- *Recommendation.* **N06** Do not defend AC generators as a bilateral priority: trusted supply is adequate (AIB LOW INTERVENTION), yet Canada's observed protection or attention is high.

11. Evidence Register

- Every recommendation (IDs J, C, T, I, X, D, N, M) has rows in evidence_register.csv: the OSI metrics behind it (variable, value for each country, year, source, evidence specificity) and any external items, with URL and access date. Recommendations that rest on external research only carry basis = external.[1]
- External evidence vintage: accessed 2026-11-06; 256 items (partial 197, unverified 18, verified 41).[1]

Where the findings are weak or counterintuitive

- Joint bets resting on a node need inherited from its cell (a proxy): Uranium conversion & enrichment; Accumulator parts & separators.
- Evidence-led compacts (OSI cannot see the partner's capability in trade data): none.
- Canada has no listed trade position (under 1% of world supply) at nodes where Korea has material demand: wafers, polysilicon, synthetic graphite (anode-grade), rare-earth oxides & compounds, electric motors, rare-earth permanent magnets, lithium chemicals, pv cover glass, refrigerants, battery materials (foil, electrolyte), battery recycling feed, graphite electrodes, semiconductor manufacturing equipment, silver powder, direct-reduced iron (dri/hbi). Trade data miss capability that is owned abroad or sold at home, so these nodes cannot show complementarity even where corporate evidence suggests it.
- Low-evidence units among the active states: none.
- Fuel-trade lanes reach tier 1 (oil extraction; coal components; coal assembly; buyer: Korea and Canada): real energy-security lanes, but outside the electro-industrial compacts and with no market failure beyond the supply concentration the engine measures.

Sources

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Appendix — evidence tables and method

The tables below carry the full complementarity map, the node profiles and the trade actions that the main text summarises. Method, decision rules and limitations: docs/partnership_memos.md. Machine-readable outputs: country_profiles.csv, dyad_complementarity.csv, compact_candidates.csv, trade_priorities.csv, investment_priorities.csv, third_country_opportunities.csv, deal_pipeline.csv and evidence_register.csv.

Table A1. Section 2, What Each Country Is Good At — and What It Needs

COMPACT	NODE	STAGE	KOREA SHARE	KOREA NEED	CANADA SHARE	CANADA NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-BATTERIES	Natural graphite	Extraction	2.6%	high 1.00	3.4%	low (cell 0.96; not a material buyer)	BUILD	CHN 82%
SC-BATTERIES	Lithium minerals	Extraction	0.3%	high 1.00	3.1%	low (cell 0.96; not a material buyer)	LOW INTERVENTION	CHN 25%
SC-BATTERIES	Battery recycling feed	Recovery	0.3%	high 0.94	0.1%	high 0.85	LOW INTERVENTION	CHN 8%
SC-BATTERIES	Nickel intermediates & chemicals	Refining	1.0%	low (cell 0.99; not a material buyer)	10.2%	low 0.47	LOW INTERVENTION	CHN 37%
SC-BATTERIES	Cobalt intermediates & chemicals	Refining	0.8%	high 0.99	3.5%	low 0.47	BUILD	CHN 76%
SC-BATTERIES	Lithium chemicals	Refining	0.7%	high 0.99	0.0%	low 0.47	BUILD	CHN 74%
SC-BATTERIES	Accumulator parts & separators	Components	12.7%	low (cell 0.97; not a material buyer)	0.1%	high 0.71	BUILD	CHN 38%
SC-BATTERIES	Cathode materials & precursors	Refining	50.9%	low (cell 0.99; not a material buyer)	1.1%	low 0.47	BUILD	CHN 92%
SC-BATTERIES	Synthetic graphite (anode-grade)	Refining	4.6%	high 0.99	0.0%	low 0.47	BUILD	CHN 94%
SC-BATTERIES	Battery materials (foil, electrolyte)	Refining	9.5%	high 0.99	0.0%	low 0.47	BUILD	CHN 16%
SC-BATTERIES	Li-ion cells & packs	Assembly	4.0%	low (cell 0.97; not a material buyer)	0.7%	low (cell 0.68; not a material buyer)	BUILD	CHN 84%
SC-BATTERIES	Other accumulators	Assembly	1.1%	low (cell 0.83; not a material buyer)	0.2%	high 0.68	BUILD	CHN 28%
SC-EV	Electric buses, trucks & tractors	Assembly	0.4%	demand 0.37	6.6%	demand 0.62	BUILD	CHN 24%
SC-EV	Electric & plug-in cars	Assembly	7.0%	low 0.37	0.4%	demand 0.62	BUILD	CHN 69%
SC-EV	Electric two-wheelers & works trucks	Assembly	1.1%	low 0.37	1.5%	demand 0.62	BUILD	CHN 30%
SC-EV	Electric rail vehicles	Assembly	10.6%	low n/a	0.0%	low n/a	LOW INTERVENTION	CHN 14%
SC-GRID	Copper ores & concentrates	Extraction	0.0%	high 0.97	3.0%	low (cell 0.80; not a material buyer)	LOW INTERVENTION	CHN 13%
SC-GRID	Copper smelting & refining	Refining	1.7%	high 0.96	1.3%	low 0.53	LOW INTERVENTION	CHN 50%
SC-GRID	Aluminium wire rod	Refining	0.0%	not measured n/a	12.7%	low n/a	LOW INTERVENTION	RUS 7%
SC-GRID	Grain-oriented electrical steel (GOES)	Refining	5.2%	low (cell 0.96; not a material buyer)	0.1%	demand 0.53	BUILD	CHN 41%

COMPACT	NODE	STAGE	KOREA SHARE	KOREA NEED	CANADA SHARE	CANADA NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-GRID	Copper wire & rod	Refining	1.3%	low (cell 0.96; not a material buyer)	6.5%	low (cell 0.69; not a material buyer)	LOW INTERVENTION	CHN 6%
SC-GRID	Insulators & bushings	Components	0.8%	low (cell 0.80; not a material buyer)	1.6%	demand 0.50	BUILD	CHN 32%
SC-GRID	Power cables	Components	8.4%	low 0.64	1.1%	demand 0.62	LOW INTERVENTION	CHN 19%
SC-GRID	Electricity meters	Assembly	0.0%	low 0.64	1.6%	demand 0.49	BUILD	CHN 34%
SC-GRID	Transformers	Assembly	9.8%	low 0.64	2.9%	low 0.49	LOW INTERVENTION	CHN 21%
SC-GRID	Switchgear & protection	Assembly	3.7%	low 0.64	1.4%	demand 0.62	LOW INTERVENTION	CHN 20%
SC-HEAT	Refrigerants	Refining	0.1%	demand n/a	0.0%	demand n/a	BUILD	CHN 45%
SC-HEAT	Refrigeration & heat-pump compressors	Components	7.0%	low n/a	0.1%	demand n/a	BUILD	CHN 33%
SC-HEAT	Heat pumps & AC units	Assembly	1.9%	low n/a	1.7%	demand n/a	BUILD	CHN 49%
SC-INDUSTRY	PGMs (iridium, ruthenium)	Refining	0.0%	not measured n/a	5.8%	low n/a	LOW INTERVENTION	RUS 30%
SC-INDUSTRY	Electrolysers	Assembly	5.7%	low n/a	1.6%	low n/a	BUILD	CHN 32%
SC-INDUSTRY	Hydrogen, ammonia & methanol	Refining	0.2%	demand n/a	4.3%	low n/a	LOW INTERVENTION	RUS 3%
SC-INDUSTRY	Furnace parts	Components	2.2%	low n/a	1.0%	low n/a	BUILD	CHN 21%
SC-INDUSTRY	Graphite electrodes	Components	0.0%	demand n/a	0.1%	demand n/a	COORDINATE	CHN 38%
SC-INDUSTRY	Direct-reduced iron (DRI/HBI)	Refining	0.0%	demand n/a	0.0%	not measured n/a	LOW INTERVENTION	RUS 20%
SC-INDUSTRY	Electric furnaces & heating	Assembly	5.3%	low n/a	0.6%	demand n/a	BUILD	CHN 27%
SC-MAGNETS	Rare-earth oxides & compounds	Refining	1.0%	high 0.99	0.0%	low (cell 0.94; not a material buyer)	BUILD	CHN 84%
SC-MAGNETS	Non-oriented electrical steel (NGO)	Refining	13.7%	low (cell 0.96; not a material buyer)	0.0%	demand 0.53	BUILD	CHN 26%
SC-MAGNETS	Rare-earth permanent magnets	Components	0.9%	high 0.97	0.0%	low 0.58	BUILD	CHN 64%
SC-MAGNETS	Ferrite & other magnets	Components	4.4%	low (cell 0.97; not a material buyer)	0.2%	demand 0.58	BUILD	CHN 58%
SC-MAGNETS	Electric motors	Assembly	2.6%	high 0.97	0.8%	low 0.58	BUILD	CHN 26%
SC-MAGNETS	AC generators	Assembly	0.3%	high 0.80	1.0%	low 0.62	LOW INTERVENTION	CHN 15%
SC-NUCLEAR	Uranium/fuel Extraction: aggregated low-coverage sub-sectors (Spent fuel & nuclear scrap, Uranium ores)	Extraction	0.0%	low (cell 0.96; not a material buyer)	23.1%	low 0.61	INSUFFICIENT DATA	RUS 8%
SC-NUCLEAR	Uranium conversion & enrichment	Refining	0.0%	high 0.98	15.9%	low 0.61	LOW INTERVENTION	RUS 16%
SC-NUCLEAR	Fuel assemblies & zirconium	Components	7.4%	low 0.54	0.0%	low (cell 0.74; not a material buyer)	LOW INTERVENTION	RUS 28%

COMPACT	NODE	STAGE	KOREA SHARE	KOREA NEED	CANADA SHARE	CANADA NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-NUCLEAR	Nuclear special materials	Refining	0.1%	demand n/a	41.5%	low n/a	LOW INTERVENTION	0%
SC-SEMIS	Gallium, germanium & indium	Refining	18.0%	low n/a	2.6%	low n/a	BUILD	CHN 25%
SC-SEMIS	Semiconductor manufacturing equipment	Assembly	5.6%	demand 0.25	0.0%	low 0.36	LOW INTERVENTION	CHN 5%
SC-SEMIS	Power semiconductors	Components	3.2%	low (cell 0.70; not a material buyer)	0.0%	low 0.50	BUILD	CHN 32%
SC-SEMIS	Sensors & controller parts	Components	2.4%	low n/a	0.8%	low n/a	BUILD	CHN 24%
SC-SEMIS	Integrated circuits	Components	17.1%	low n/a	0.0%	not measured n/a	LOW INTERVENTION	CHN 16%
SC-SEMIS	Power electronics components	Components	2.5%	low (cell 0.97; not a material buyer)	1.9%	low 0.58	BUILD	CHN 37%
SC-SEMIS	Automatic controllers	Assembly	2.8%	demand 0.51	2.8%	low 0.48	LOW INTERVENTION	CHN 12%
SC-SOLAR	Polysilicon	Refining	1.2%	high 0.71	0.0%	low (cell 0.67; not a material buyer)	LOW INTERVENTION	CHN 97%
SC-SOLAR	Wafers	Refining	6.4%	demand 0.30	0.0%	low 0.50	BUILD	CHN 98%
SC-SOLAR	Silver powder	Refining	1.6%	demand n/a	0.1%	low n/a	LOW INTERVENTION	0%
SC-SOLAR	Solar cells	Components	11.1%	low 0.30	0.0%	low 0.50	BUILD	CHN 92%
SC-SOLAR	PV cover glass	Components	1.9%	demand 0.30	0.8%	low 0.50	BUILD	CHN 61%
SC-WIND	Wind blades & hubs	Components	1.1%	low (cell 0.80; not a material buyer)	2.8%	low 0.45	BUILD	CHN 71%
SC-WIND	Wind towers	Components	11.7%	low (cell 0.80; not a material buyer)	1.1%	low 0.45	LOW INTERVENTION	CHN 18%
SC-WIND	Hydro turbine parts	Components	0.0%	not measured n/a	2.2%	low n/a	COORDINATE	CHN 35%
SC-WIND	Wind turbine generating sets	Assembly	0.1%	low 0.64	0.0%	demand 0.49	BUILD	CHN 74%

Sources: 1.

Table A2. Section 3, Bilateral Complementarity Map

RANK	TIER	TECHNOLOGY	UNIT	STAGE	KOREA POSITION	CANADA POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
1	joint bet	Electric Vehicles	Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds)	Refining	weak	moderate	Shared vulnerability	geopolitical chokepoint; need 0.99 / 0.94
2	joint bet	Wind	Wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds)	Refining	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.75 / 0.69
3	joint bet	Batteries	Batteries extraction (cobalt ores, lithium minerals, natural graphite)	Extraction	weak	moderate	Capacity-constrained trusted supply	geopolitical chokepoint; need 1.00 / 0.96
4	joint bet	Nuclear	Nuclear refining (uranium conversion & enrichment)	Refining	weak	moderate	Canada supplies / Korea needs	strategic manufacturing gap; need 0.98 / 0.61
5	joint bet	Nuclear Fuel & Reactor Systems	Uranium conversion & enrichment	Refining	weak (0.0% of world supply)	strong (15.9% of world supply)	Canada supplies / Korea needs	strategic manufacturing gap; need 0.98 / 0.61
6	joint bet	Nuclear	Nuclear extraction (uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores))	Extraction	weak	strong	Canada supplies / Korea needs	geopolitical chokepoint; need 0.96 / 0.61
7	joint bet	Nuclear	Nuclear components (fuel assemblies & zirconium, reactor parts)	Components	strong	moderate	Korea supplies / Canada needs	strategic manufacturing gap; need 0.54 / 0.74

RANK	TIER	TECHNOLOGY	UNIT	STAGE	KOREA POSITION	CANADA POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
8	joint bet	Nuclear	Nuclear assembly (nuclear reactors & enrichment machinery)	Assembly	strong	moderate	Korea supplies / Canada needs	strategic manufacturing gap; need 0.62 / 0.67
9	joint bet	Solar	Solar refining (copper wire & rod, polysilicon)	Refining	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.71 / 0.67
10	joint bet	Battery Materials & Cells	Accumulator parts & separators	Components	strong (12.7% of world supply)	weak (0.1% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.97 / 0.71
11	joint bet	Oil	Oil extraction	Extraction	weak	strong	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.99 / 0.84
12	joint bet	Coal	Coal components	Components	strong	moderate	Korea supplies / Canada needs	strategic manufacturing gap; need 0.58 / 0.81
13	joint bet	Batteries	Batteries components (accumulator parts & separators)	Components	moderate	moderate	Shared vulnerability	strategic manufacturing gap; need 0.73 / 0.71
14	joint bet	Coal	Coal assembly	Assembly	strong	moderate	Korea supplies / Canada needs	strategic manufacturing gap; need 0.60 / 0.68
15	coordinate	Battery Materials & Cells	Cathode materials & precursors	Refining	strong (50.9% of world supply)	moderate (1.1% of world supply)	Complementary specialisation	geopolitical chokepoint; need 0.99 / 0.47
16	coordinate	Electric Vehicles	Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite)	Extraction	weak	moderate	Capacity-constrained trusted supply	geopolitical chokepoint; need 1.00 / 0.96
17	coordinate	Permanent Magnets & Drive Systems	Non-oriented electrical steel (NGO)	Refining	strong (13.7% of world supply)	weak (0.0% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.96 / 0.53
18	coordinate	Battery Materials & Cells	Nickel intermediates & chemicals	Refining	weak (1.0% of world supply)	strong (10.2% of world supply)	Complementary specialisation	strategic manufacturing gap; need 0.99 / 0.47
19	coordinate	Grid Equipment & Conductors	Power cables	Components	strong (8.4% of world supply)	weak (1.1% of world supply)	Complementary specialisation	strategic manufacturing gap; need 0.64 / 0.62
20	coordinate	Oil	Oil assembly (ac generators)	Assembly	strong	moderate	Complementary specialisation	strategic manufacturing gap; need 0.30 / 0.62
21	coordinate	Grid Equipment & Conductors	Aluminium wire rod	Refining	weak (0.0% of world supply)	strong (12.7% of world supply)	Complementary specialisation	strategic manufacturing gap; need n/a / n/a
22	coordinate	Coal	Coal extraction	Extraction	weak	moderate	Canada supplies / Korea needs	no material intervention; need 0.94 / 0.54
23	manage competition	Batteries	Batteries assembly (li-ion cells & packs, other accumulators)	Assembly	strong	strong	Direct competition	geopolitical chokepoint; need 0.83 / 0.68

Sources: 1.

Table A3. Section 4, Trade Agenda

ID	DIRECTION	UNIT	ACTION TYPE	ACTION
T01	both	Electric Vehicles/Refining	anti circumvention	Common origin and anti-circumvention rules for Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds): both import heavily from CHN, so a measure one side takes alone leaks through the other.
T02	both	Wind/Refining	common standards	Common specifications for Wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds) so a joint supplier base can serve both markets.
T03	CAN -> KOR	Batteries/Extraction	procurement recognition	Recognise Canada output of Batteries extraction (cobalt ores, lithium minerals, natural graphite) as trusted domestic-equivalent supply in Korea procurement and content rules; capacity, not access, is the constraint.
T04	CAN -> KOR	Nuclear/Refining	predictability	Keep Canada exports of Nuclear refining (uranium conversion & enrichment) to Korea tariff-free and predictable; the buyer's need is met by the trusted supplier.
T05	CAN -> KOR	uranium-conversion-and-enrichment	predictability	Keep Canada exports of Uranium conversion & enrichment to Korea tariff-free and predictable; the buyer's need is met by the trusted supplier.

ID	DIRECTION	UNIT	ACTION TYPE	ACTION
T06	CAN -> KOR	Nuclear/Extraction	predictability	Keep Canada exports of Nuclear extraction (uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores)) to Korea tariff-free and predictable; the buyer's need is met by the trusted supplier.
T07	KOR -> CAN	Nuclear/Components	predictability	Keep Korea exports of Nuclear components (fuel assemblies & zirconium, reactor parts) to Canada tariff-free and predictable; the buyer's need is met by the trusted supplier.
T08	KOR -> CAN	Nuclear/Assembly	predictability	Keep Korea exports of Nuclear assembly (nuclear reactors & enrichment machinery) to Canada tariff-free and predictable; the buyer's need is met by the trusted supplier.
T09	both	Solar/Refining	common standards	Common specifications for Solar refining (copper wire & rod, polysilicon) so a joint supplier base can serve both markets.
T10	KOR -> CAN	accumulator-parts-and-separators	procurement recognition	Recognise Korea output of Accumulator parts & separators as trusted domestic-equivalent supply in Canada procurement and content rules; capacity, not access, is the constraint.
T11	CAN -> KOR	Oil/Extraction	procurement recognition	Recognise Canada output of Oil extraction as trusted domestic-equivalent supply in Korea procurement and content rules; capacity, not access, is the constraint.
T12	KOR -> CAN	Coal/Components	predictability	Keep Korea exports of Coal components to Canada tariff-free and predictable; the buyer's need is met by the trusted supplier.
T13	open	nuclear-special-materials	open sourcing	Leave Nuclear special materials open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T14	open	battery-recycling-feed	open sourcing	Leave Battery recycling feed open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T15	open	semiconductor-manufacturing-equipment	open sourcing	Leave Semiconductor manufacturing equipment open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T16	open	copper-ores-and-concentrates	open sourcing	Leave Copper ores & concentrates open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T17	open	lithium-minerals	open sourcing	Leave Lithium minerals open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T18	open	silver-powder	open sourcing	Leave Silver powder open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T19	both	Batteries/Assembly	no preference	No bilateral preference for Batteries assembly (li-ion cells & packs, other accumulators): both are strong suppliers; keep the market contestable.

Sources: 1.

Rule-generated from the data (R/partnerships) with external facts from a dated, curated evidence layer; not yet checked by an independent verifier. Items marked partial or unverified must be confirmed before citation.
Allied Industrial Base Explorer, Partnership memo: Korea–Canada Strategic Energy & Industrial Partnership